

Additional Problems on Linear Equations and Their Graphs (Homework)

For each problem below, write an equation of the line with the given characteristics.

- ① X-intercept: 2
y-intercept: 4

Answers
 $y = -2x + 4$
Other forms are possible.

- ② X-intercept: -3
y-intercept: 5

$$y = \frac{5}{3}x + 5$$

- ③ X-intercept: 18
y-intercept: -6

$$y = \frac{1}{3}x - 6$$

- ④ X-intercept: -7
y-intercept: -1

$$y = -\frac{1}{7}x - 1$$

- ⑤ Slope: -4
y-intercept: 9

$$y = -4x + 9$$

- ⑥ X-intercept: 4
point: (3, 7)

$$y - 7 = -7(x - 3)$$

- ⑦ Slope: 3
X-intercept: -2

$$y = 3x + 6$$

- ⑧ y-intercept: 1
point: (-6, -3)

$$y = \frac{2}{3}x + 1$$

- ⑨ Point: (-4, -6)
Slope: $-\frac{3}{2}$

$$y + 6 = -\frac{3}{2}(x + 4)$$

- ⑩ y-intercept 8 and
parallel to $y = 4x + 2$.

$$y = 4x + 8$$

- ⑪ y-intercept -3 and
perpendicular to $y = -\frac{5}{4}x + 7$.

$$y = \frac{4}{5}x - 3$$

- ⑫ y-intercept 5 and
perpendicular to $y = 3x + 1$.

$$y = -\frac{1}{3}x + 5$$

- ⑬ Parallel to $y = 2x - 2$ and
containing the point (4, 3).

$$y - 3 = 2(x - 4)$$

- ⑭ Perpendicular to $y = 7x - \frac{1}{2}$
and containing the point (-1, -6).

$$y + 6 = -\frac{1}{7}(x + 1)$$

- ⑮ Perpendicular to $y - 4 = -\frac{1}{2}(x + 1)$
and containing the point (-5, 3).

$$y - 3 = 2(x + 5)$$

- ⑯ Parallel to $y + 3 = -7(x + 8)$
and containing the point (2, 1).

$$y - 1 = -7(x - 2)$$

- ⑰ y-intercept 1 and
parallel to $2x + 5y = 5$.

$$y = -\frac{2}{5}x + 1$$

- ⑱ y-intercept -3 and
perpendicular to $-4x + 2y = 7$.

$$y = -\frac{1}{2}x - 3$$

- ⑲ Perpendicular to $3x + 4y = 5$
and containing the point (2, -7).

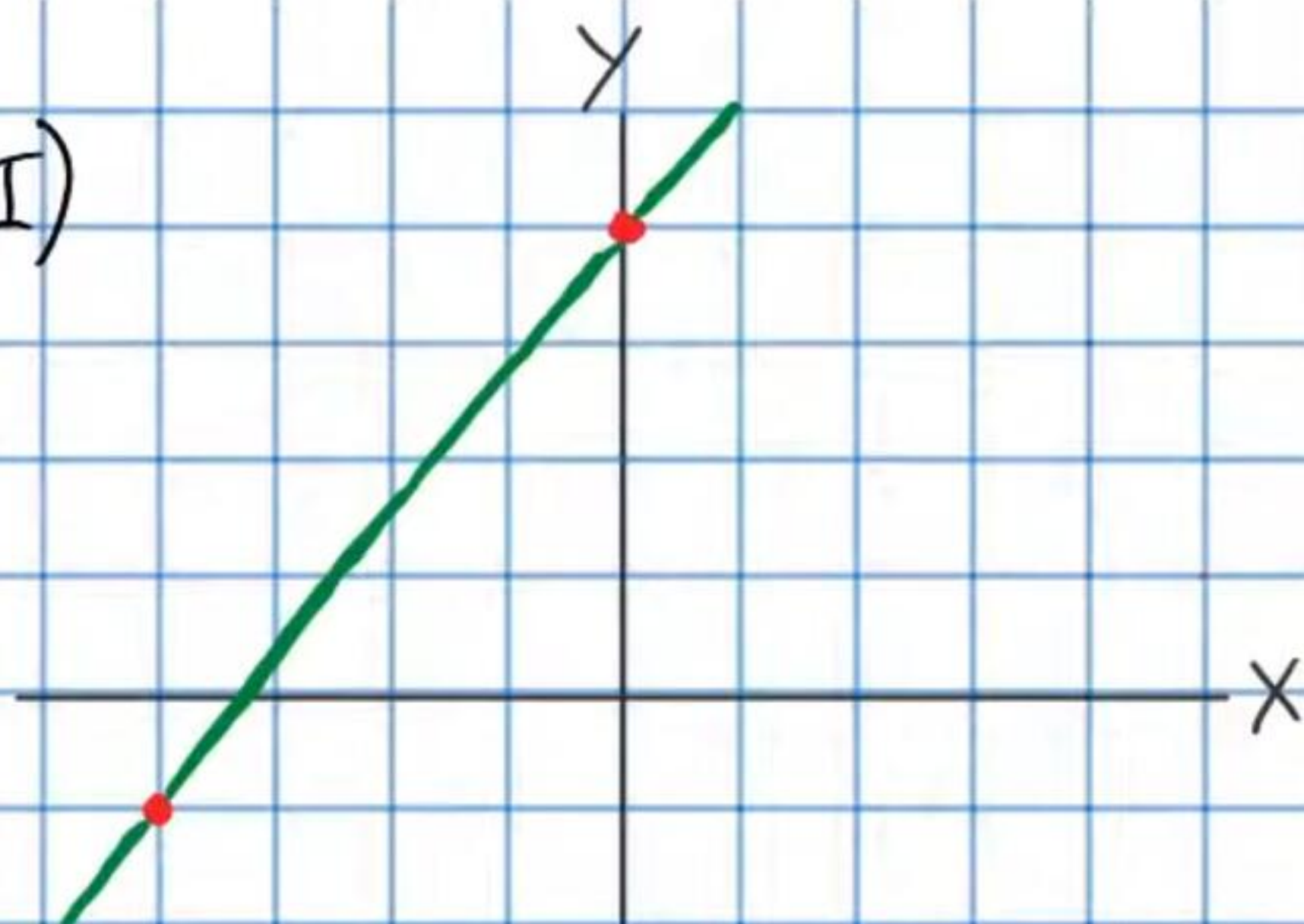
$$y + 7 = \frac{4}{3}(x - 2)$$

- ⑳ Parallel to $-6x - 4y = 16$
and containing the point (-1, -1).

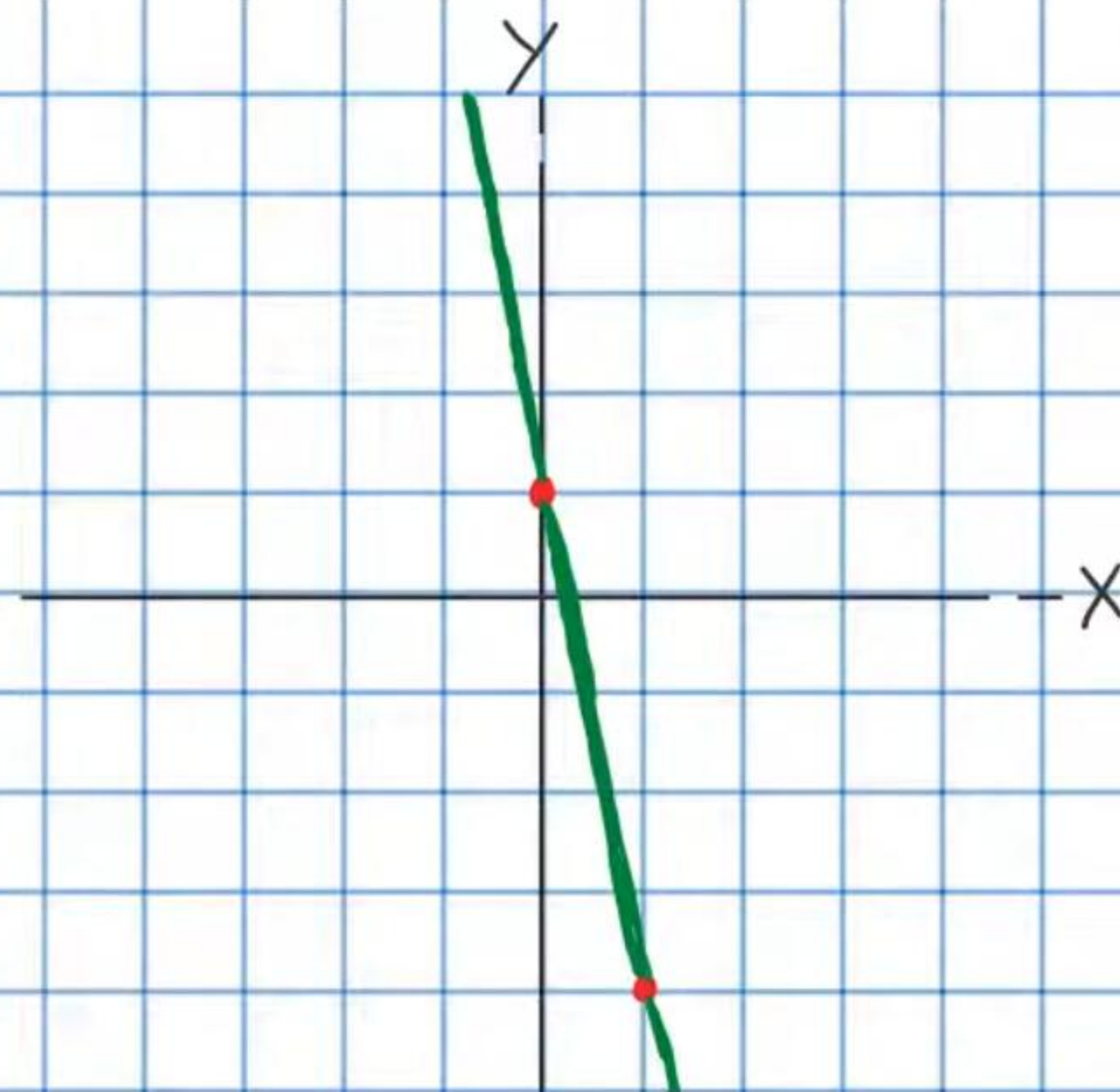
$$y + 1 = -\frac{3}{2}(x + 1)$$

For each graph below, write an equation of the line shown.

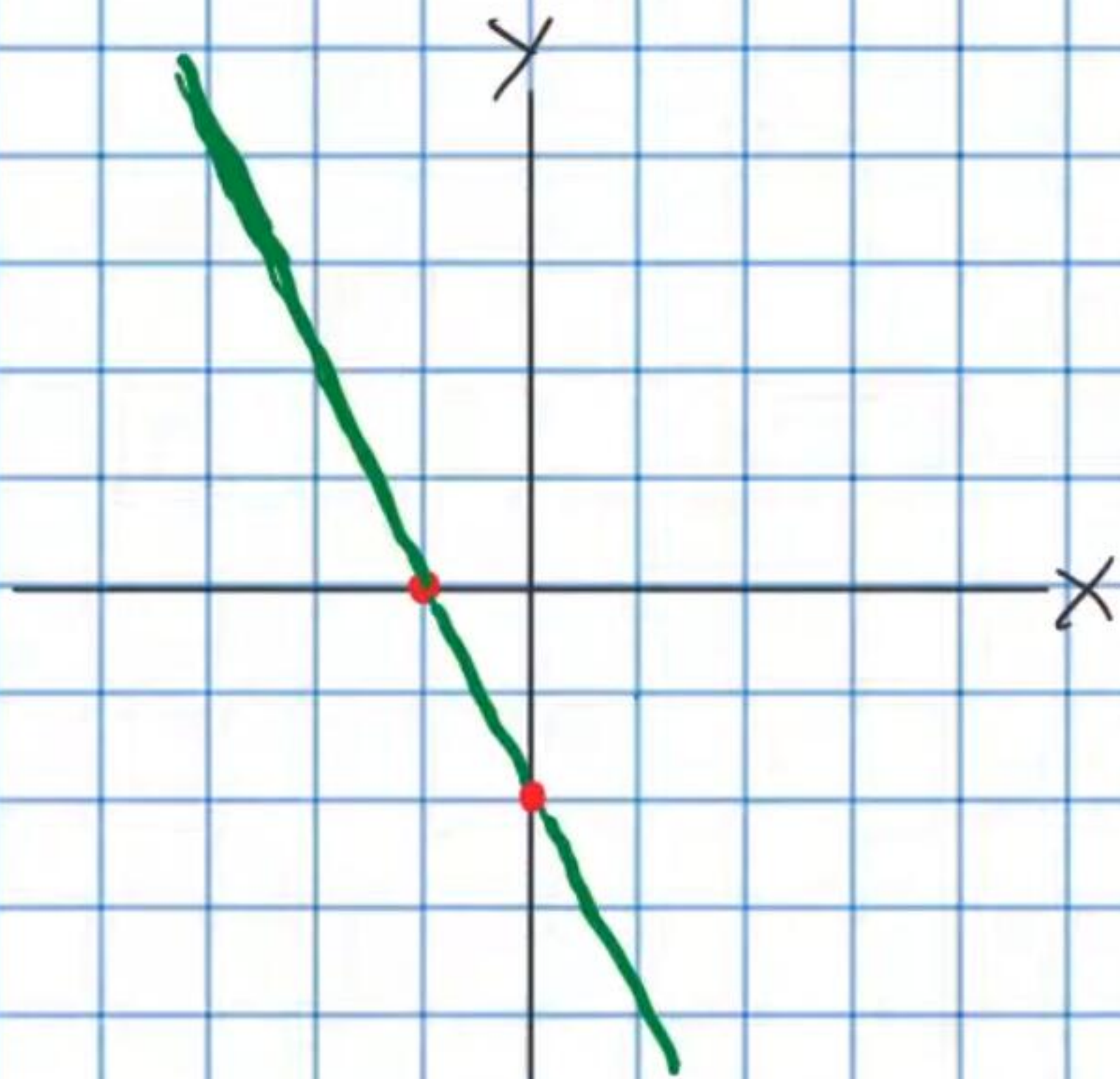
② I)



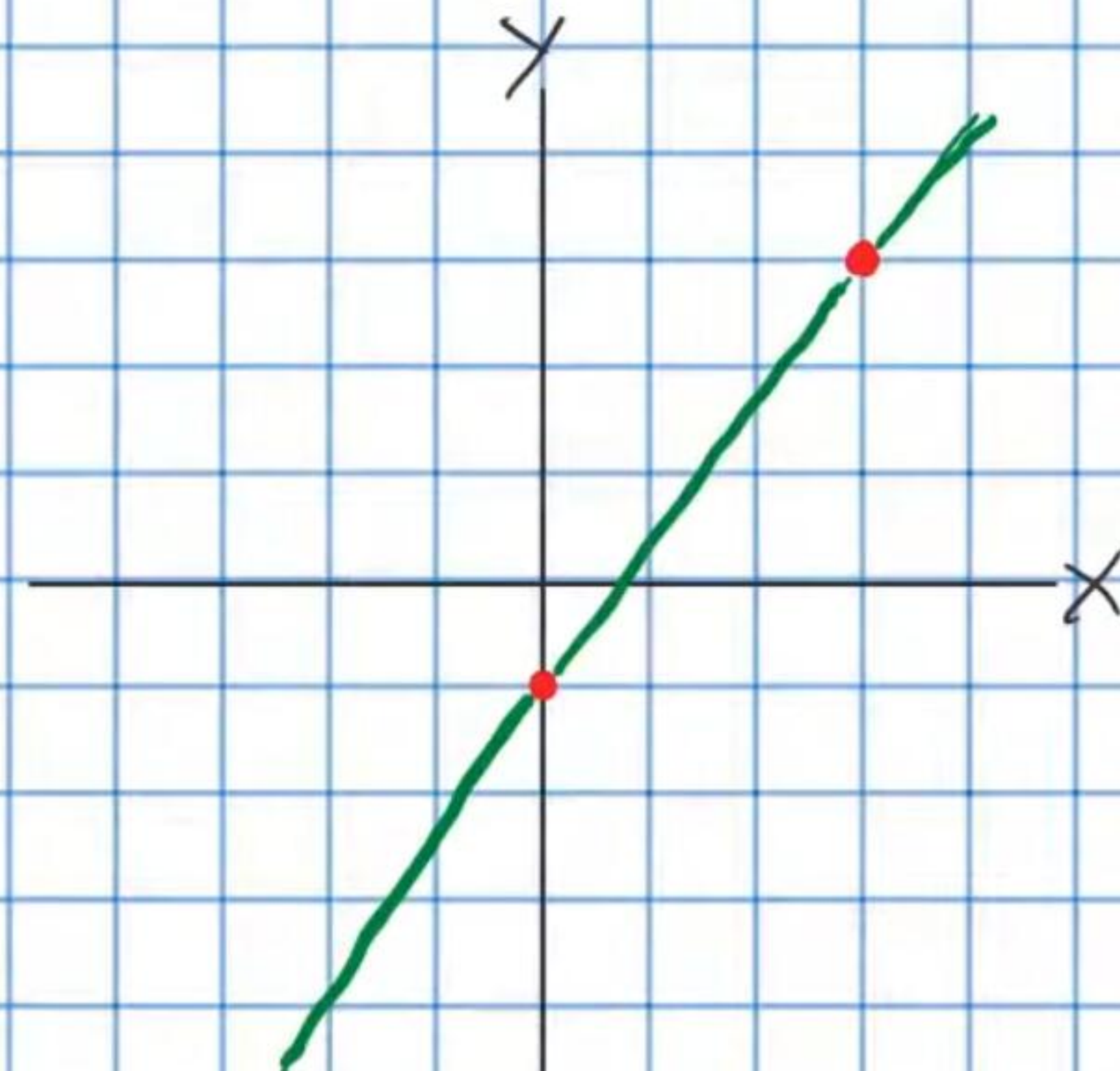
II)



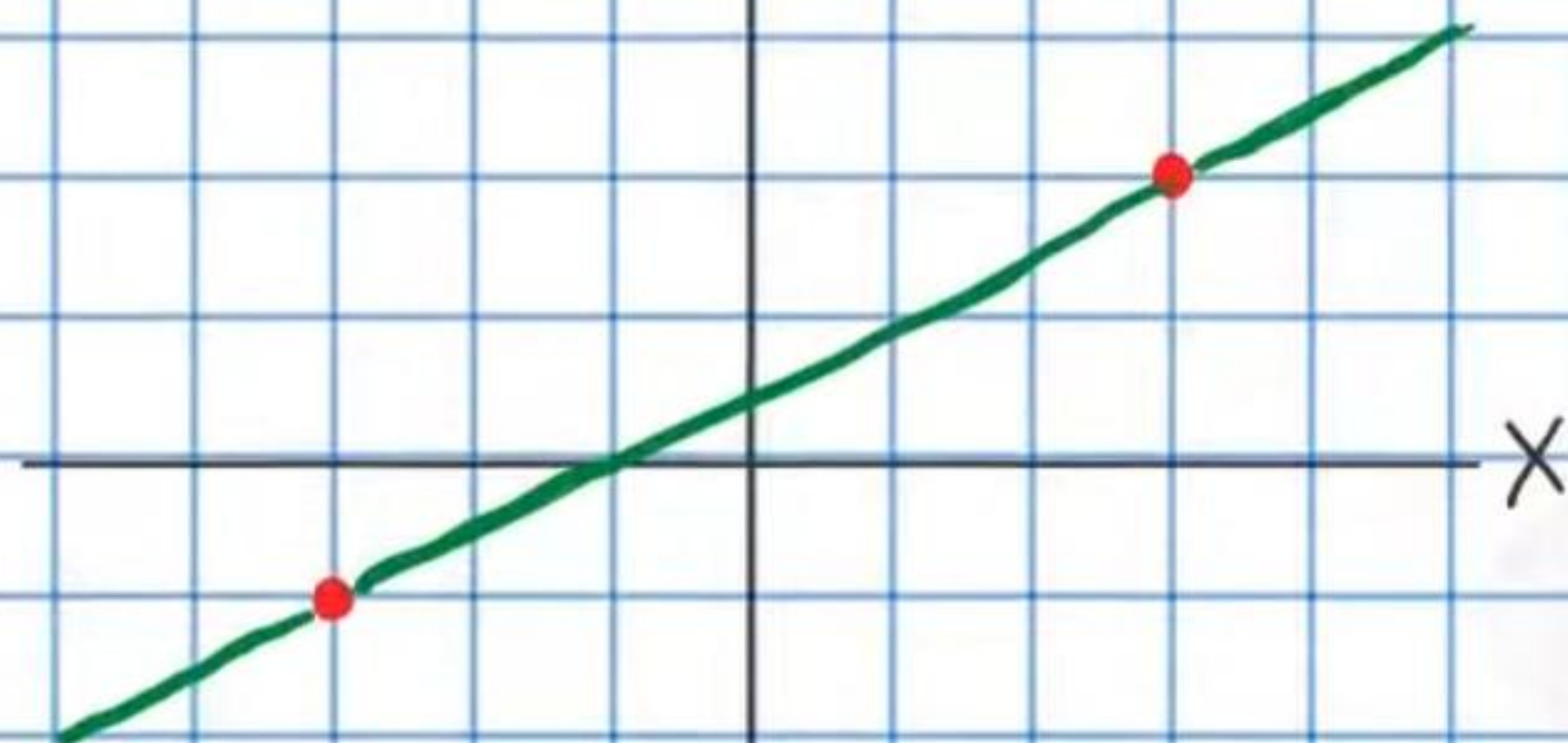
② I)



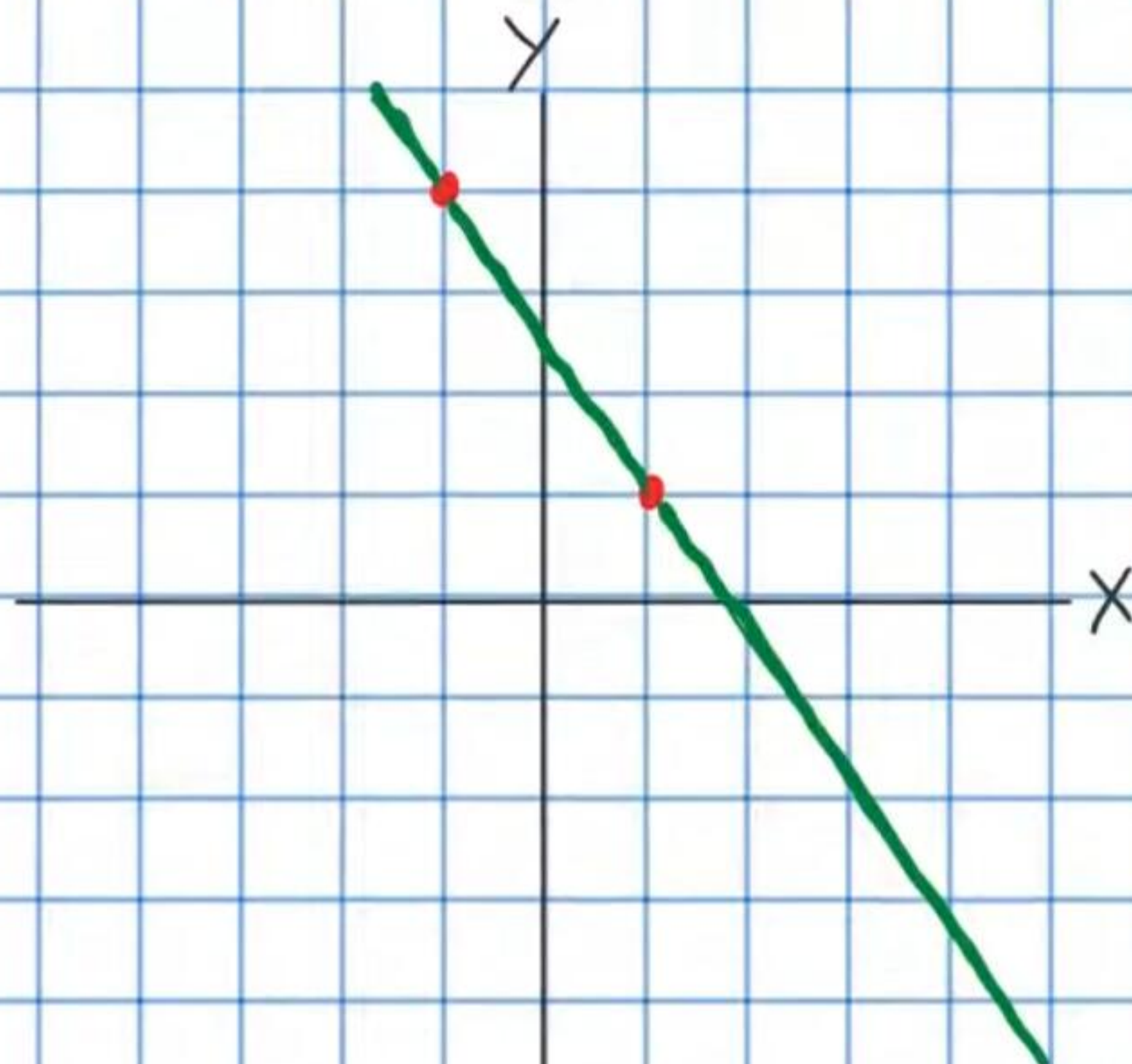
II)



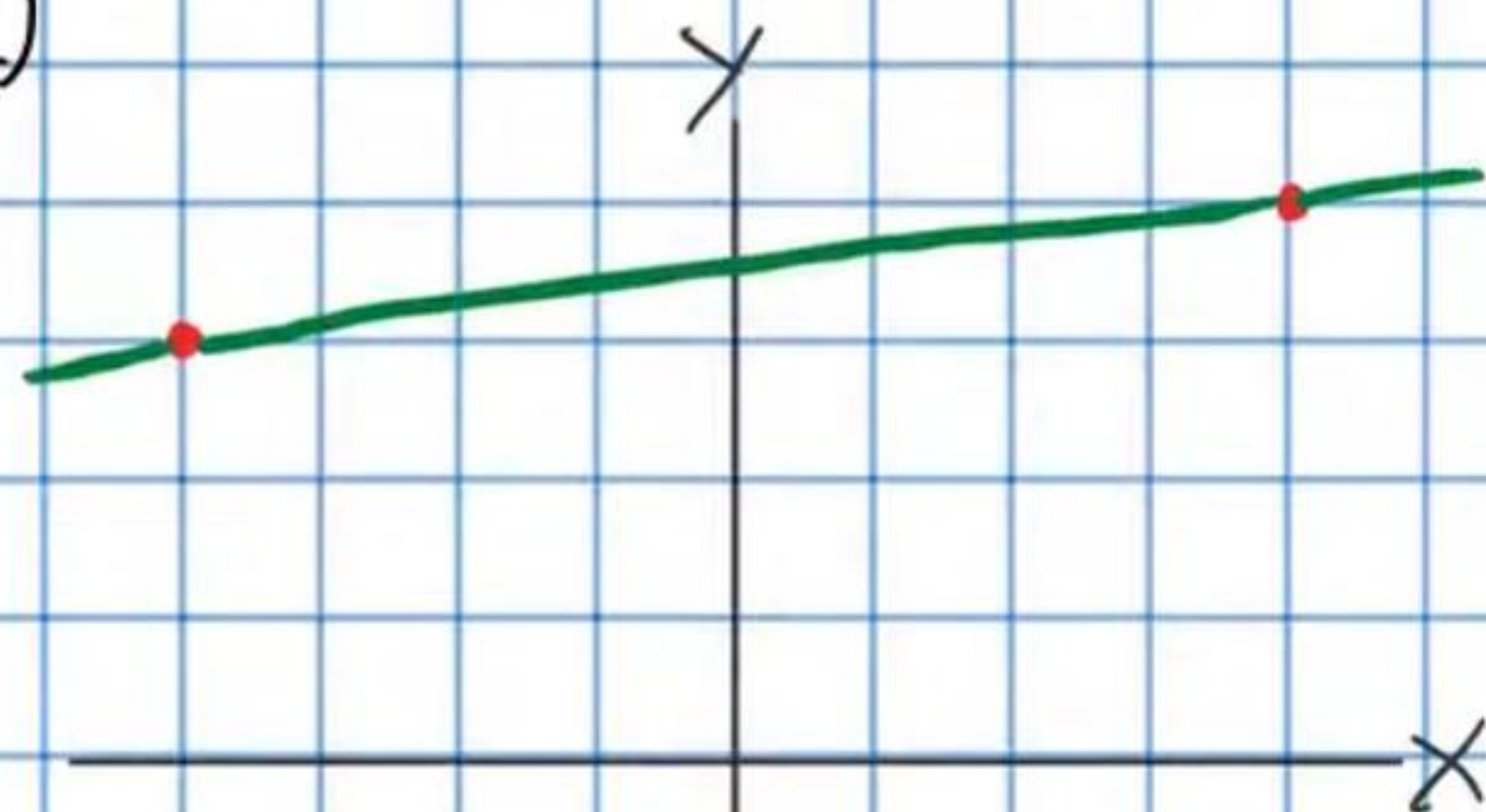
23 I)



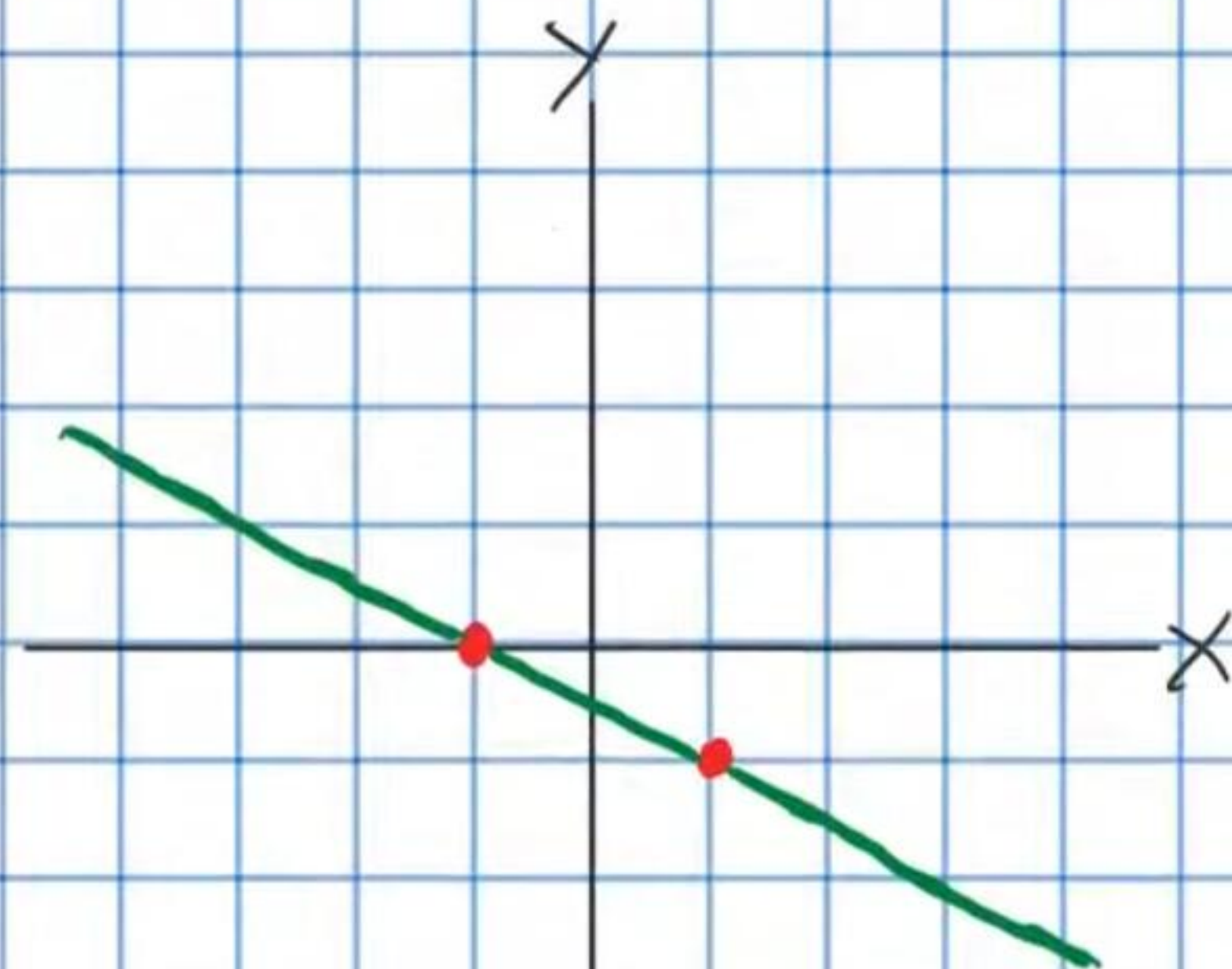
II)



24 I)

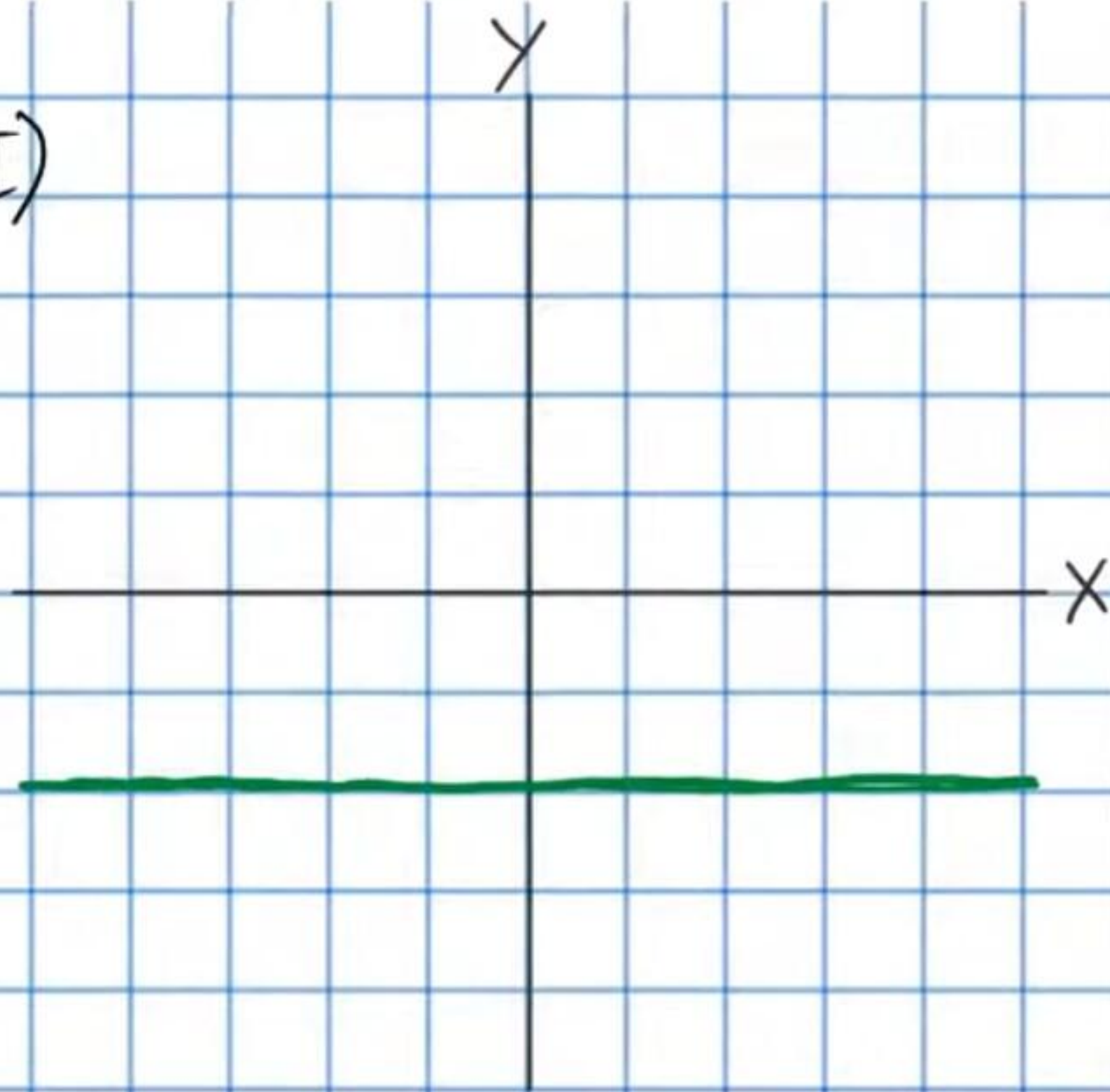


II)

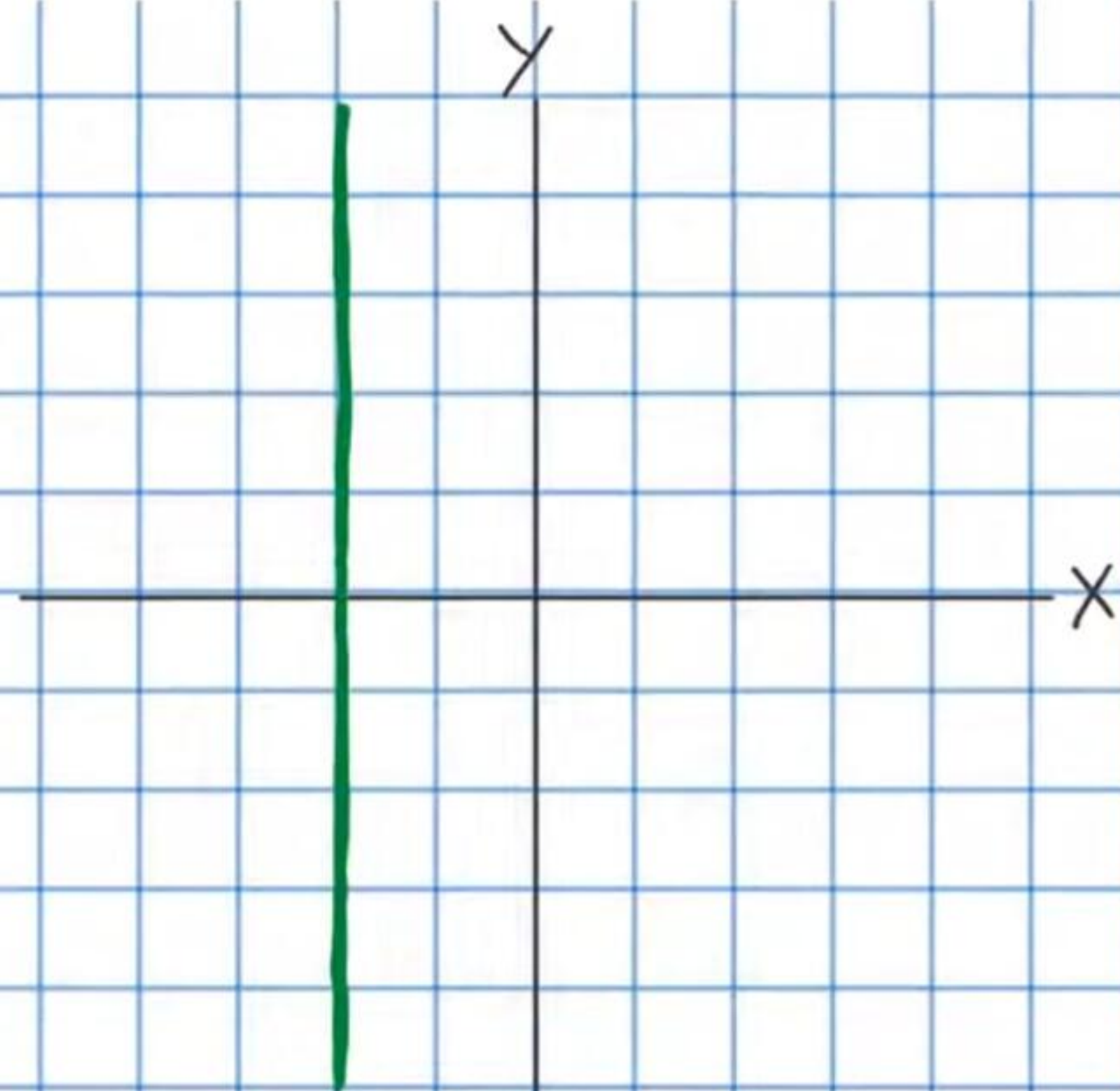


For each graph below, write an equation of the line shown and identify its slope.

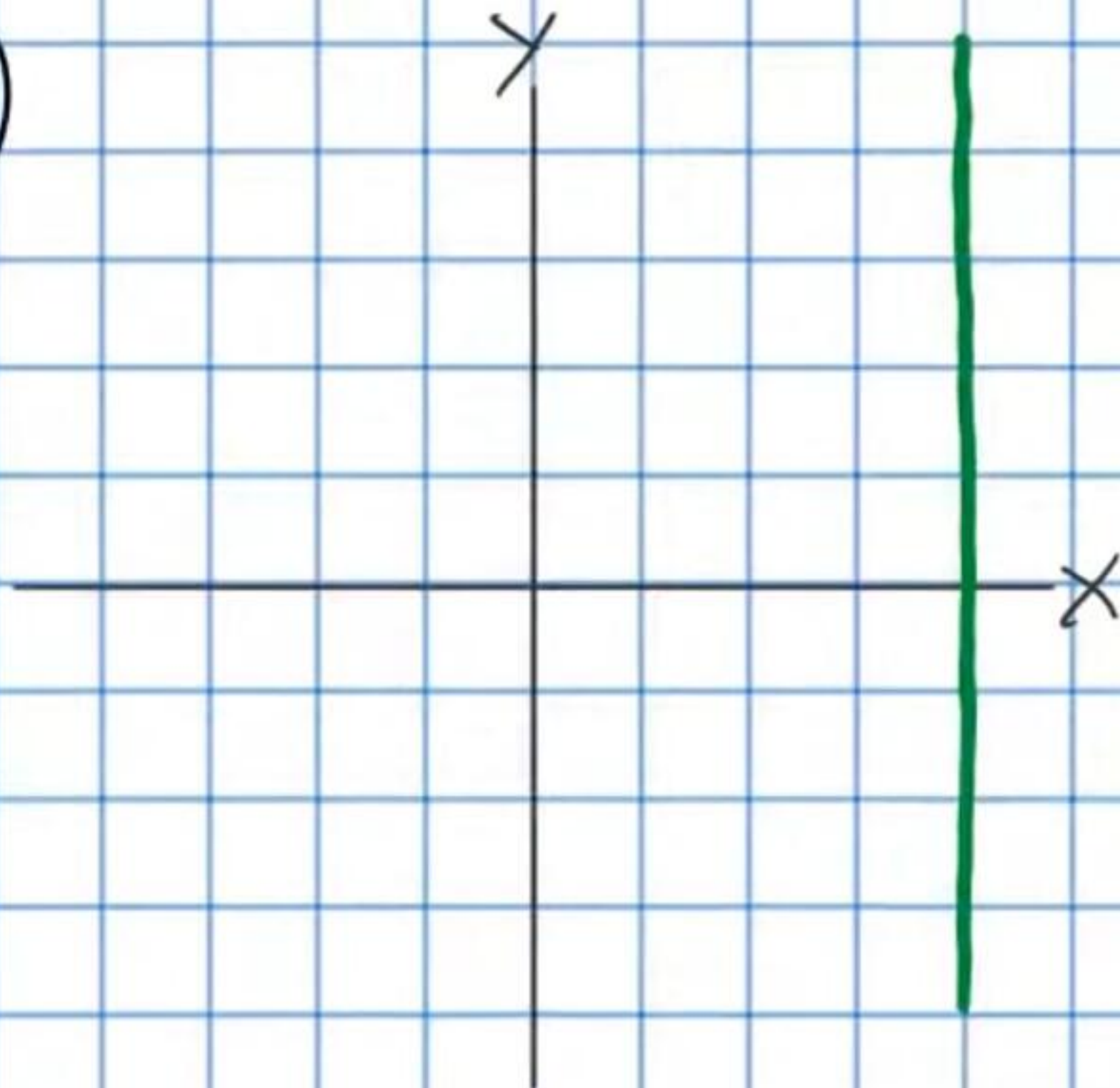
(25) I)



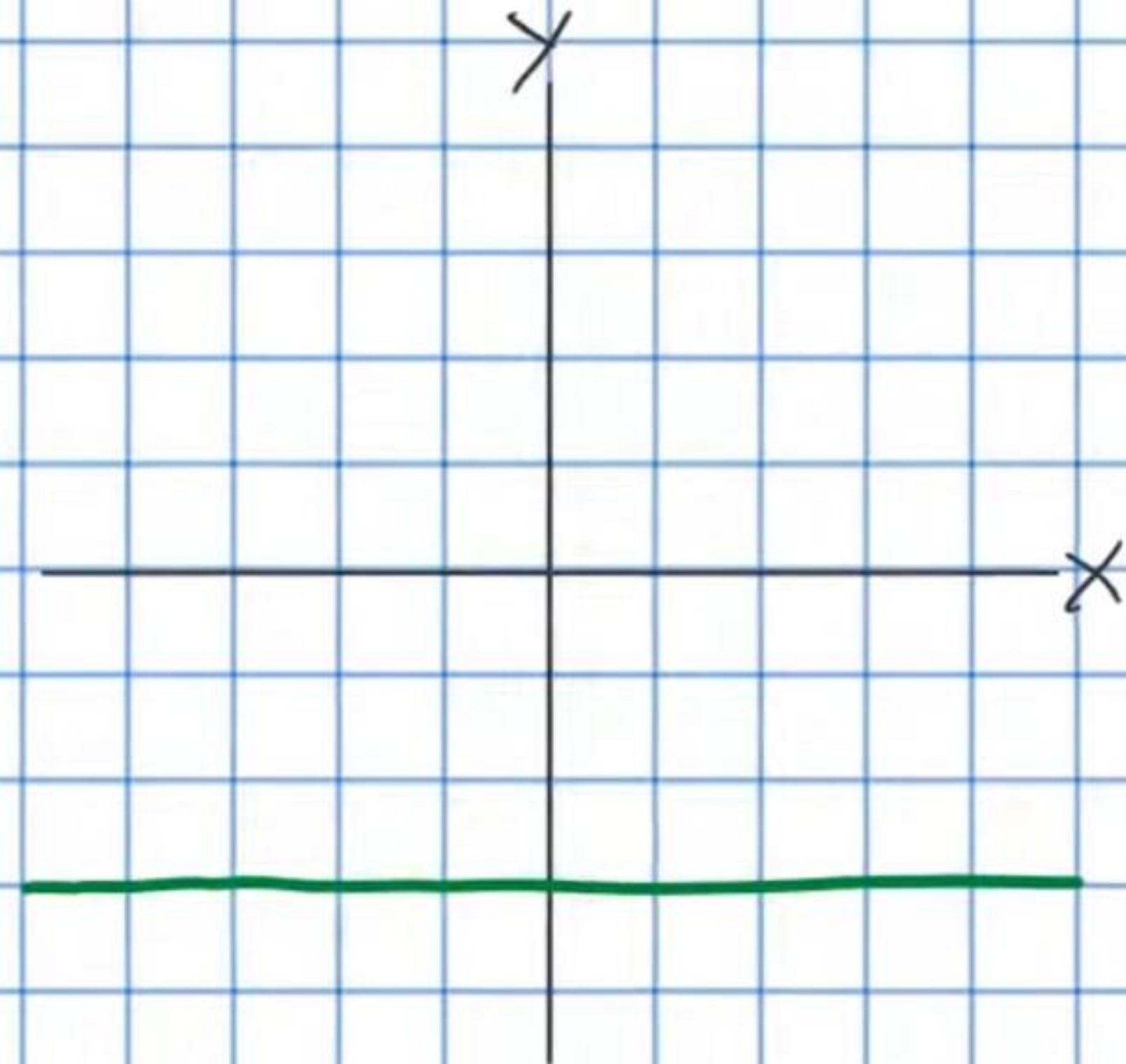
II)



(26) I)

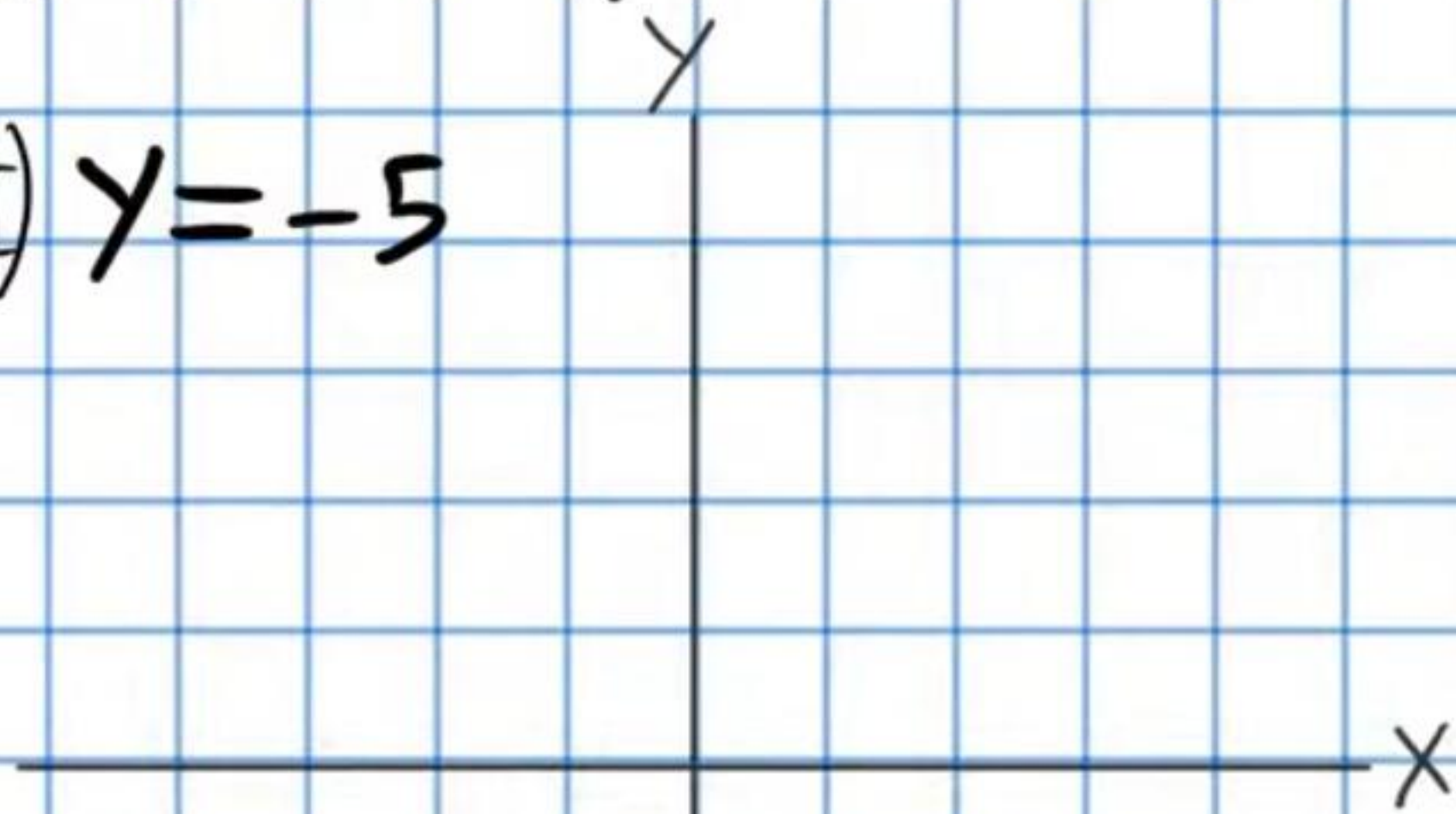


II)

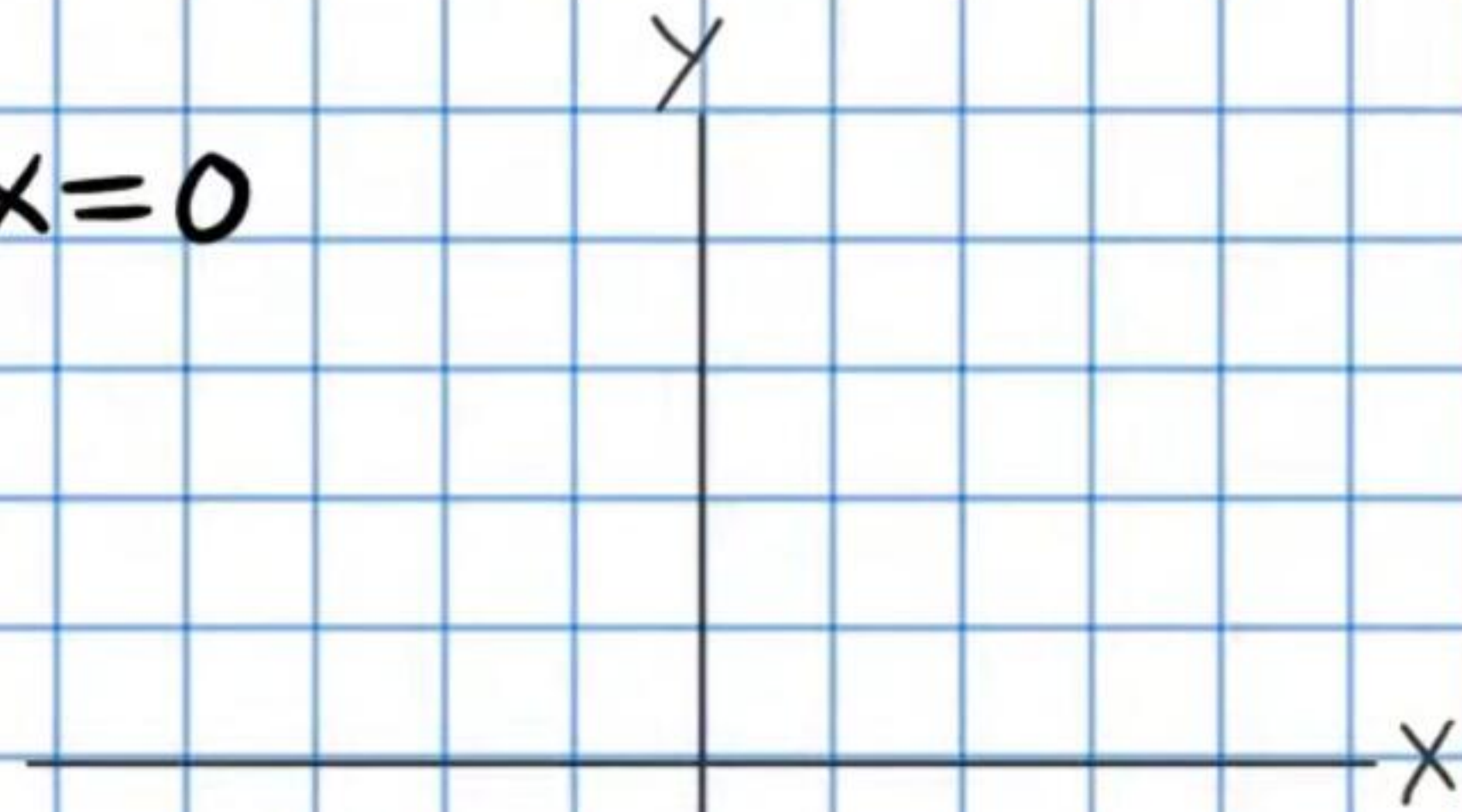


Graph each equation below and identify the slope of the line produced.

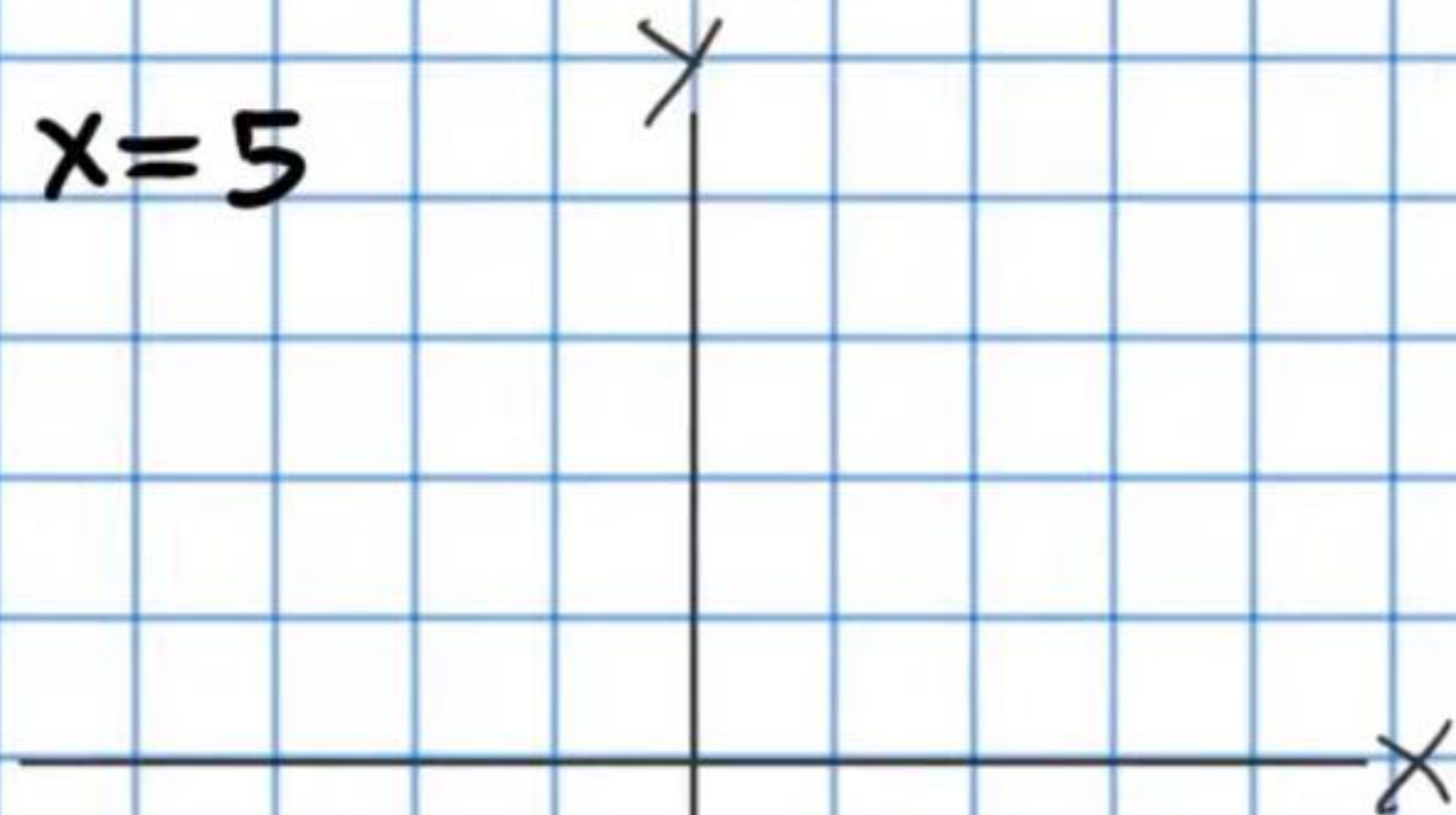
②⑦ I) $y = -5$



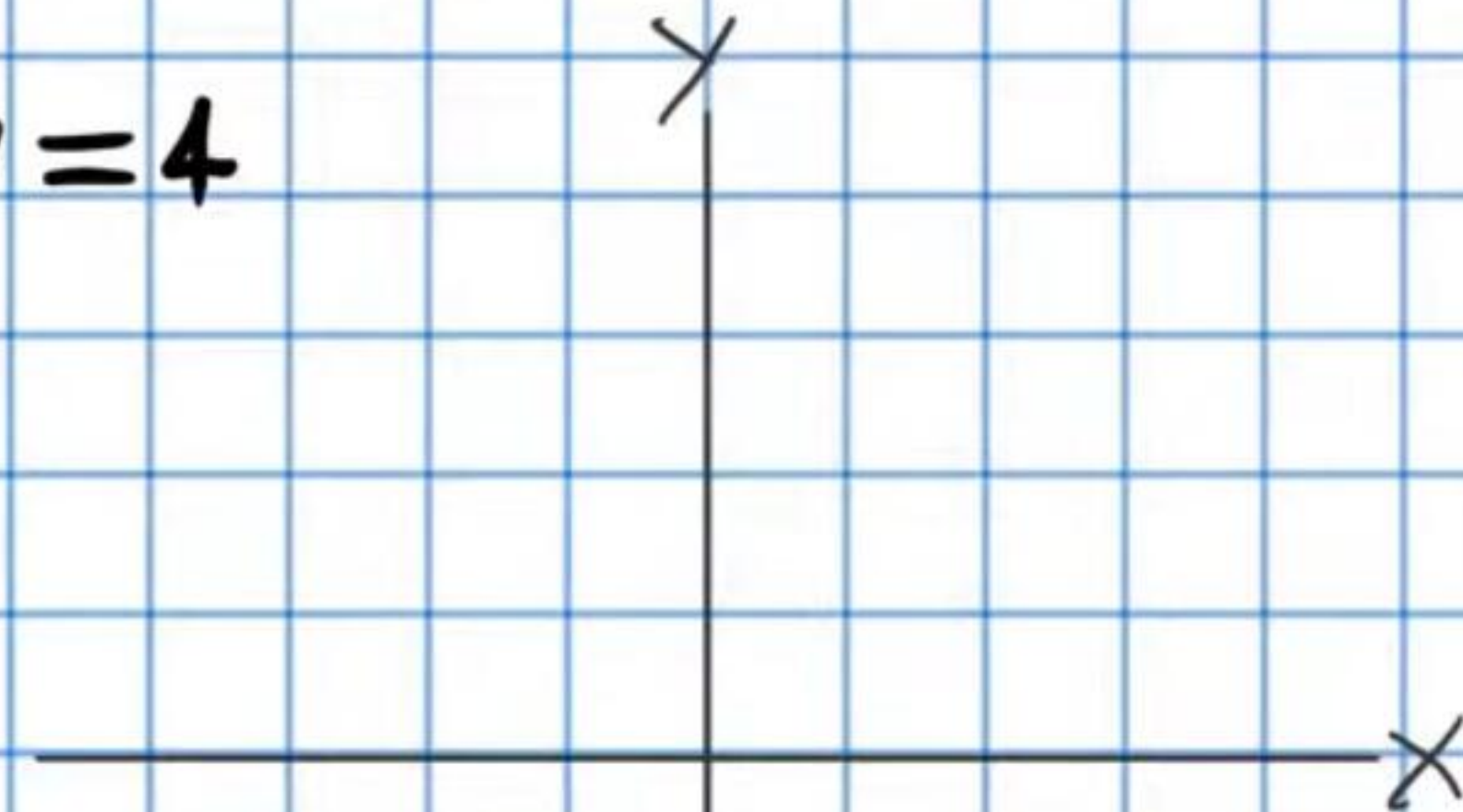
II) $x = 0$



②⑧ I) $x = 5$



II) $y = 4$



Answers for Problem #'s 21-28

★ Other forms are possible, but these are easiest to find with the given information.

②① I) $y = \frac{5}{4}x + 4$

II) $y = -5x + 1$

②② I) $y = -2x - 2$

II) $y = \frac{4}{3}x - 1$

②③ I) $y + 1 = \frac{1}{2}(x + 3)$ or $y - 2 = \frac{1}{2}(x - 3)$

II) $y - 4 = -\frac{3}{2}(x + 1)$ or $y - 1 = -\frac{3}{2}(x - 1)$

②④ I) $y - 3 = \frac{1}{8}(x + 4)$ or $y - 4 = \frac{1}{8}(x - 4)$

II) $y - 0 = -\frac{1}{2}(x + 1)$ or $y + 1 = -\frac{1}{2}(x - 1)$

②⑤ I) $y = -2$ $m = 0$

II) $x = -2$ $m = \text{und}$

②⑥ I) $x = 4$ $m = \text{und}$

II) $y = -3$ $m = 0$

②⑦ I) $y = -5$

II) $x = 0$

$m = 0$

$m = \text{und}$

②⑧ I) $x = 5$

II) $y = 4$

$m = 0$

$m = \text{und}$